# Apply Filters to SQL Queries

## Project Description
As a security professional at a large organization, I investigated potential security issues involving login attempts and employee machines. Using SQL filters with AND, OR, and NOT operators, I queried the organization's database to retrieve specific records from the `employees` and `log_in_attempts` tables. This helped identify suspicious login activity and isolate employee data by department for further security investigation.

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## Retrieve After Hours Failed Login Attempts

A potential security incident occurred after business hours (after 18:00).
I needed to retrieve all failed login attempts that occurred after this time.

```sql
SELECT *
FROM log_in_attempts
WHERE login_time > '18:00' AND success = FALSE;
```

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00' AND success = FALSE;
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  |       0 |
|       20 | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  |       0 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   |       0 |
|       34 | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   |       0 |
|       42 | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   |       0 |
|       52 | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   |       0 |
|       69 | wjaffrey | 2022-05-11 | 19:55:15   | USA     | 192.168.100.17  |       0 |
|       82 | abernard | 2022-05-12 | 23:38:46   | MEX     | 192.168.234.49  |       0 |
|       87 | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 |       0 |
|       96 | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  |       0 |
|      104 | asundara | 2022-05-11 | 18:38:07   | US      | 192.168.96.200  |       0 |
|      107 | bisles   | 2022-05-12 | 20:25:57   | USA     | 192.168.116.187 |       0 |
|      111 | aestrada | 2022-05-10 | 22:00:26   | MEXICO  | 192.168.76.27   |       0 |
|      127 | abellmas | 2022-05-09 | 21:20:51   | CANADA  | 192.168.70.122  |       0 |
|      131 | bisles   | 2022-05-09 | 20:03:55   | US      | 192.168.113.171 |       0 |
|      155 | cgriffin | 2022-05-12 | 22:18:42   | USA     | 192.168.236.176 |       0 |
|      160 | jclark   | 2022-05-10 | 20:49:00   | CANADA  | 192.168.214.49  |       0 |
|      199 | yappiah  | 2022-05-11 | 19:34:48   | MEXICO  | 192.168.44.232  |       0 |
+----------+----------+------------+------------+---------+-----------------+---------+
19 rows in set (0.001 sec)

MariaDB [organization]>
```

**Explanation:**
The `WHERE` clause filters records using two conditions joined by `AND`:
- `login_time > '18:00'` retrieves only attempts that occurred after 6:00 PM
- `success = 0` retrieves only failed login attempts (0 = false/failed)

Both conditions must be true for a record to be returned, ensuring only
After-hours failed attempts are shown.

---
## Retrieve Login Attempts on Specific Dates

A suspicious event was flagged on 2022-05-09. I needed to investigate all
login attempts on that date and the day before (2022-05-08).

```sql
SELECT *
FROM log_in_attempts
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
+----------+----------+------------+------------+---------+----------------+---------+
| event_id | username | login_date | login_time | country | ip_address     | success |
+----------+----------+------------+------------+---------+----------------+---------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |       1 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |       1 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |       0 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |       1 |
|       15 | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  |       0 |
|       24 | arusso   | 2022-05-09 | 06:49:39   | MEXICO  | 192.168.171.192 |       1 |
|       25 | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  |       1 |
|       26 | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 |       1 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   |       0 |
|       30 | yappiah  | 2022-05-09 | 03:22:22   | MEX     | 192.168.124.48  |       1 |
|       32 | acook    | 2022-05-09 | 02:52:02   | CANADA  | 192.168.142.239 |       0 |
|       36 | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  |       1 |
|       38 | sbaelish | 2022-05-09 | 14:40:01   | USA     | 192.168.60.42   |       1 |
|       39 | yappiah  | 2022-05-09 | 07:56:40   | MEXICO  | 192.168.57.115  |       1 |
|       42 | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   |       0 |
|       43 | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  |       0 |
|       44 | daquino  | 2022-05-08 | 07:02:35   | CANADA  | 192.168.168.144 |       0 |
|       47 | dkot     | 2022-05-08 | 05:06:45   | US      | 192.168.233.24  |       1 |
|       49 | asundara | 2022-05-08 | 14:00:01   | US      | 192.168.173.213 |       0 |
|       53 | nmason   | 2022-05-08 | 11:51:38   | CAN     | 192.168.133.188 |       1 |
|       56 | acook    | 2022-05-08 | 04:56:30   | CAN     | 192.168.209.130 |       1 |
|       58 | ivelasco | 2022-05-09 | 17:20:54   | CAN     | 192.168.57.162  |       0 |
|       61 | dtanaka  | 2022-05-09 | 09:45:18   | USA     | 192.168.98.221  |       1 |
|       65 | aalonso  | 2022-05-09 | 23:42:12   | MEX     | 192.168.52.37   |       1 |
|       66 | aestrada | 2022-05-08 | 21:58:32   | MEX     | 192.168.67.223  |       1 |
|       67 | abernard | 2022-05-09 | 11:53:41   | MEX     | 192.168.118.29  |       1 |
|       68 | mrah     | 2022-05-08 | 17:16:13   | US      | 192.168.42.248  |       1 |
|       70 | tmitchel | 2022-05-09 | 10:55:17   | MEXICO  | 192.168.87.199  |       1 |
|       71 | mcouliba | 2022-05-09 | 06:57:42   | CAN     | 192.168.55.169  |       0 |
|       72 | alevitsk | 2022-05-08 | 12:09:10   | CANADA  | 192.168.139.176 |       1 |
|       79 | abernard | 2022-05-09 | 11:41:15   | MEX     | 192.168.158.170 |       0 |
|       80 | cjackson | 2022-05-08 | 02:18:10   | CANADA  | 192.168.33.140  |       1 |
|       83 | lrodriqu | 2022-05-08 | 08:10:23   | USA     | 192.168.67.69   |       1 |
|       87 | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 |       0 |
|       90 | gesparza | 2022-05-09 | 00:49:05   | CANADA  | 192.168.87.201  |       0 |
|       92 | pwashing | 2022-05-08 | 00:36:12   | US      | 192.168.247.219 |       0 |
|       96 | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  |       0 |
|       97 | jreckley | 2022-05-09 | 02:49:23   | MEXICO  | 192.168.32.231  |       1 |
|      101 | sbaelish | 2022-05-08 | 12:01:22   | US      | 192.168.145.158 |       0 |
```

| | | | | | | |
|---|---|---|---|---|---|---|
| 83 | lrodriqu | 2022-05-08 | 08:10:23 | USA | 192.168.67.69 | 1 |
| 87 | apatel | 2022-05-08 | 22:38:31 | CANADA | 192.168.132.153 | 0 |
| 90 | gesparza | 2022-05-09 | 00:49:05 | CANADA | 192.168.87.201 | 0 |
| 92 | pwashing | 2022-05-08 | 00:36:12 | US | 192.168.247.219 | 0 |
| 96 | ivelasco | 2022-05-09 | 22:36:36 | CAN | 192.168.84.194 | 0 |
| 97 | jreckley | 2022-05-09 | 02:49:23 | MEXICO | 192.168.32.231 | 1 |
| 101 | sbaelish | 2022-05-08 | 12:01:22 | US | 192.168.145.158 | 0 |
| 102 | jreckley | 2022-05-09 | 16:51:44 | MEX | 192.168.108.13 | 1 |
| 108 | daquino | 2022-05-09 | 21:30:48 | CANADA | 192.168.15.110 | 1 |
| 110 | mabadi | 2022-05-09 | 00:01:54 | USA | 192.168.90.124 | 1 |
| 112 | rjensen | 2022-05-09 | 09:22:05 | MEX | 192.168.69.116 | 1 |
| 117 | bsand | 2022-05-08 | 00:19:11 | USA | 192.168.197.187 | 0 |
| 120 | tmitchel | 2022-05-09 | 02:58:17 | MEXICO | 192.168.134.62 | 0 |
| 127 | abellmas | 2022-05-09 | 21:20:51 | CANADA | 192.168.70.122 | 0 |
| 128 | jclark | 2022-05-09 | 10:45:59 | CANADA | 192.168.122.169 | 0 |
| 131 | bisles | 2022-05-09 | 20:03:55 | US | 192.168.113.171 | 0 |
| 134 | iuduike | 2022-05-09 | 06:46:40 | USA | 192.168.22.115 | 1 |
| 135 | bsand | 2022-05-09 | 14:06:33 | US | 192.168.91.238 | 0 |
| 144 | daquino | 2022-05-09 | 11:09:32 | CANADA | 192.168.139.9 | 0 |
| 145 | ivelasco | 2022-05-08 | 09:06:02 | CANADA | 192.168.39.196 | 1 |
| 147 | yappiah | 2022-05-08 | 06:04:34 | MEX | 192.168.65.245 | 0 |
| 148 | daquino | 2022-05-08 | 06:15:55 | CANADA | 192.168.135.6 | 1 |
| 150 | nmason | 2022-05-08 | 14:40:02 | CAN | 192.168.204.124 | 0 |
| 151 | mabadi | 2022-05-09 | 16:29:46 | USA | 192.168.30.225 | 1 |
| 158 | smartell | 2022-05-09 | 19:30:32 | MEXICO | 192.168.190.178 | 1 |
| 161 | abellmas | 2022-05-09 | 13:25:50 | CAN | 192.168.180.205 | 0 |
| 162 | yappiah | 2022-05-09 | 04:51:22 | MEXICO | 192.168.162.100 | 0 |
| 163 | tmitchel | 2022-05-08 | 09:21:16 | MEX | 192.168.119.29 | 0 |
| 165 | jreckley | 2022-05-08 | 15:28:43 | MEXICO | 192.168.34.193 | 0 |
| 168 | jlansky | 2022-05-08 | 13:25:42 | USA | 192.168.210.94 | 1 |
| 169 | alevitsk | 2022-05-08 | 08:10:43 | CANADA | 192.168.210.228 | 0 |
| 170 | sbaelish | 2022-05-09 | 16:43:18 | USA | 192.168.65.113 | 0 |
| 172 | mabadi | 2022-05-08 | 08:06:50 | US | 192.168.180.41 | 1 |
| 178 | sgilmore | 2022-05-08 | 12:27:22 | CAN | 192.168.52.216 | 0 |
| 184 | alevitsk | 2022-05-08 | 03:09:48 | CAN | 192.168.33.70 | 0 |
| 186 | bisles | 2022-05-09 | 04:29:17 | USA | 192.168.40.72 | 0 |
| 187 | arusso | 2022-05-09 | 00:36:26 | MEX | 192.168.77.137 | 0 |
| 189 | nmason | 2022-05-08 | 05:37:24 | CANADA | 192.168.168.117 | 1 |
| 190 | jsoto | 2022-05-09 | 05:09:21 | USA | 192.168.25.60 | 0 |
| 191 | cjackson | 2022-05-08 | 06:46:07 | CANADA | 192.168.7.187 | 0 |
| 193 | lrodriqu | 2022-05-08 | 07:11:29 | US | 192.168.125.240 | 0 |
| 197 | jsoto | 2022-05-08 | 09:05:09 | US | 192.168.36.21 | 0 |

75 rows in set (0.001 sec)

MariaDB [organization]>

**Explanation:**
The `OR` operator returns records where either condition is true:
- `login_date = '2022-05-09'` retrieves attempts on the suspicious date
- `login_date = '2022-05-08'` retrieves attempts on the day before

Using `OR` ensures all login activity across both dates is captured for investigation.

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## Retrieve Login Attempts Outside of Mexico

After investigating the data, I determined that suspicious activity did not originate from Mexico. I needed to filter out all login attempts from Mexico to focus on other locations.

```sql
SELECT *
FROM log_in_attempts
WHERE NOT country LIKE 'MEX%';
```

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE NOT country LIKE 'MEX%';
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        1 | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |       1 |
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|        3 | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |       1 |
|        4 | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |       0 |
|        5 | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  |       0 |
|        7 | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 |       1 |
|        8 | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |       0 |
|       10 | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 |       0 |
|       11 | sgilmore | 2022-05-11 | 10:16:29   | CANADA  | 192.168.140.81  |       0 |
|       12 | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |       1 |
|       13 | mrah     | 2022-05-11 | 09:29:34   | USA     | 192.168.246.135 |       1 |
|       14 | sbaelish | 2022-05-10 | 10:20:18   | US      | 192.168.16.99   |       1 |
|       15 | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  |       0 |
|       16 | mcouliba | 2022-05-11 | 06:44:22   | CAN     | 192.168.172.189 |       1 |
|       17 | pwashing | 2022-05-11 | 02:33:02   | USA     | 192.168.81.89   |       1 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  |       0 |
|       19 | jhill    | 2022-05-12 | 13:09:04   | US      | 192.168.142.245 |       1 |
|       21 | iuduike  | 2022-05-11 | 17:50:00   | US      | 192.168.131.147 |       1 |
|       25 | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  |       1 |
|       26 | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 |       1 |
|       29 | bisles   | 2022-05-11 | 01:21:22   | US      | 192.168.85.186  |       0 |
|       31 | acook    | 2022-05-12 | 17:36:45   | CANADA  | 192.168.58.232  |       0 |
|       32 | acook    | 2022-05-09 | 02:52:02   | CANADA  | 192.168.142.239 |       0 |
|       33 | zbernal  | 2022-05-11 | 02:52:10   | US      | 192.168.72.59   |       1 |
|       34 | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   |       0 |
|       36 | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  |       1 |
|       37 | eraab    | 2022-05-10 | 06:03:41   | CANADA  | 192.168.152.148 |       0 |
|       38 | sbaelish | 2022-05-09 | 14:40:01   | USA     | 192.168.60.42   |       1 |
|       41 | apatel   | 2022-05-10 | 17:39:42   | CANADA  | 192.168.46.207  |       0 |
|       42 | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   |       0 |
|       43 | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  |       0 |
|       44 | daquino  | 2022-05-08 | 07:02:35   | CANADA  | 192.168.168.144 |       0 |
|       45 | dtanaka  | 2022-05-11 | 10:28:54   | US      | 192.168.223.157 |       1 |
|       46 | eraab    | 2022-05-11 | 11:29:27   | CAN     | 192.168.24.12   |       0 |
|       47 | dkot     | 2022-05-08 | 05:06:45   | US      | 192.168.233.24  |       1 |
|       48 | asundara | 2022-05-11 | 03:18:45   | USA     | 192.168.72.10   |       1 |
|       49 | asundara | 2022-05-08 | 14:00:01   | US      | 192.168.173.213 |       0 |
|       50 | jclark   | 2022-05-10 | 10:48:02   | CANADA  | 192.168.174.117 |       0 |
|       51 | jrafael  | 2022-05-10 | 22:40:01   | CANADA  | 192.168.148.115 |       1 |
|       52 | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   |       0 |
```



```
|       184 | alevitsk  | 2022-05-08 | 03:09:48 | CAN      | 192.168.33.70   |         0 |
|       185 | jsoto     | 2022-05-10 | 13:34:58 | USA      | 192.168.151.91  |         0 |
|       186 | bisles    | 2022-05-09 | 04:29:17 | USA      | 192.168.40.72   |         0 |
|       188 | jsoto     | 2022-05-11 | 00:39:09 | USA      | 192.168.21.88   |         0 |
|       189 | nmason    | 2022-05-08 | 05:37:24 | CANADA   | 192.168.168.117 |         1 |
|       190 | jsoto     | 2022-05-09 | 05:09:21 | USA      | 192.168.25.60   |         0 |
|       191 | cjackson  | 2022-05-08 | 06:46:07 | CANADA   | 192.168.7.187   |         0 |
|       192 | bisles    | 2022-05-10 | 08:32:03 | USA      | 192.168.201.40  |         1 |
|       193 | lrodriqu  | 2022-05-08 | 07:11:29 | US       | 192.168.125.240 |         0 |
|       194 | jclark    | 2022-05-12 | 14:11:04 | CAN      | 192.168.197.247 |         0 |
|       195 | alevitsk  | 2022-05-11 | 06:59:13 | CANADA   | 192.168.236.78  |         1 |
|       196 | acook     | 2022-05-10 | 09:56:48 | CAN      | 192.168.52.90   |         0 |
|       197 | jsoto     | 2022-05-08 | 09:05:09 | US       | 192.168.36.21   |         0 |
|       200 | jclark    | 2022-05-12 | 01:11:45 | CANADA   | 192.168.91.103  |         1 |
+-----------+-----------+------------+----------+----------+-----------------+-----------+
144 rows in set (0.045 sec)

MariaDB [organization]>
```

**Explanation:**
- `NOT` excludes records that match the condition
- `LIKE 'MEX%'` matches both `MEX` and `MEXICO` entries in the dataset
- The `%` wildcard represents any number of characters after `MEX`

This query returns all login attempts from every country except Mexico.

---

## Retrieve Employees in Marketing

My team needed to update machines for employees in the Marketing department located in the East building. I queried the employees table to identify them.

```sql
SELECT *
FROM employees
WHERE department = 'Marketing' AND office LIKE 'East%';
```

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Marketing' AND office LIKE 'East%';
+-------------+-------------+----------+------------+----------+
| employee_id | device_id   | username | department | office   |
+-------------+-------------+----------+------------+----------+
|        1000 | a320b137c219 | elarson  | Marketing  | East-170 |
|        1052 | a192b174c940 | jdarosa  | Marketing  | East-195 |
|        1075 | x573y883z772 | fbautist | Marketing  | East-267 |
|        1088 | k8651965m233 | rgosh    | Marketing  | East-157 |
|        1103 | NULL         | randerss | Marketing  | East-460 |
|        1156 | a184b775c707 | dellery  | Marketing  | East-417 |
|        1163 | h679i515j339 | cwilliam | Marketing  | East-216 |
+-------------+-------------+----------+------------+----------+
7 rows in set (0.001 sec)

MariaDB [organization]>
```

**Explanation:**
- `department = 'Marketing'` filters for Marketing department employees only
- `AND office LIKE 'East%'` further filters for offices in the East building
- The `%` wildcard captures all East building office variations (East-170, East-320, etc.)

Both conditions must be met for a record to be returned.

---

## Retrieve Employees in Finance or Sales

Machines for employees in the Finance and Sales departments also needed security updates. I retrieved all employees from either department.

```sql
SELECT *
FROM employees
WHERE department = 'Finance' OR department = 'Sales';
```

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Finance' OR department = 'Sales';
```

| employee_id | device_id | username | department | office |
|---|---|---|---|---|
| 1003 | d394e816f943 | sgilmore | Finance | South-153 |
| 1007 | h174i497j413 | wjaffrey | Finance | North-406 |
| 1008 | i858j583k571 | abernard | Finance | South-170 |
| 1009 | NULL | lrodriqu | Sales | South-134 |
| 1010 | k2421212m542 | jlansky | Finance | South-109 |
| 1011 | l748m120n401 | drosas | Sales | South-292 |
| 1015 | p611q262r945 | jsoto | Finance | North-271 |
| 1017 | r550s824t230 | jclark | Finance | North-188 |
| 1018 | s310t540u653 | abellmas | Finance | North-403 |
| 1022 | w237x430y567 | arusso | Finance | West-465 |
| 1024 | y976z753a267 | iuduike | Sales | South-215 |
| 1025 | z381a365b233 | jhill | Sales | North-115 |
| 1029 | d336e475f676 | ivelasco | Finance | East-156 |
| 1035 | j236k303l245 | bisles | Sales | South-171 |
| 1039 | n253o917p623 | cjackson | Sales | East-378 |
| 1041 | p929q222r778 | cgriffin | Sales | North-208 |
| 1044 | s429t157u159 | tbarnes | Finance | West-415 |
| 1045 | t567u844v434 | pwashing | Finance | East-115 |
| 1046 | u429v921w138 | daquino | Finance | West-280 |
| 1047 | v109w587x644 | cward | Finance | West-373 |
| 1048 | w167x592y375 | tmitchel | Finance | South-288 |
| 1049 | NULL | jreckley | Finance | Central-295 |
| 1050 | y132z930a114 | csimmons | Finance | North-468 |
| 1057 | f370g535h632 | mscott | Sales | South-270 |
| 1062 | k3671639m697 | redwards | Finance | North-180 |
| 1063 | l686m140n569 | lpope | Sales | East-226 |
| 1066 | o678p794q957 | ttyrell | Sales | Central-444 |
| 1069 | NULL | jpark | Finance | East-110 |
| 1071 | t244u829v723 | zdutchma | Sales | West-348 |
| 1072 | u905v920w694 | esmith | Sales | East-421 |
| 1076 | y347z204a710 | fgarcia | Finance | Central-270 |
| 1078 | a667b270c984 | sharley | Sales | North-418 |
| 1081 | d647e310f618 | qcorbit | Finance | South-290 |
| 1083 | f840g812h544 | gkoshi | Finance | West-165 |
| 1085 | h339i498j269 | cperez | Sales | East-325 |
| 1086 | i281j129k749 | lmajumda | Sales | West-499 |
| 1089 | l358m929n154 | jpark2 | Sales | West-251 |
| 1091 | n378o313p469 | rtran | Sales | Central-230 |
| 1092 | o391p779q935 | lpark | Sales | West-227 |
| 1098 | u671v146w618 | tarchamb | Sales | North-423 |

| | | | | |
|---|---|---|---|---|
| 1072 | u905v920w694 | esmith | Sales | East-421 |
| 1076 | y347z204a710 | fgarcia | Finance | Central-270 |
| 1078 | a667b270c984 | sharley | Sales | North-418 |
| 1081 | d647e310f618 | qcorbit | Finance | South-290 |
| 1083 | f840g812h544 | gkoshi | Finance | West-165 |
| 1085 | h339i498j269 | cperez | Sales | East-325 |
| 1086 | i281j129k749 | lmajumda | Sales | West-499 |
| 1089 | l358m929n154 | jpark2 | Sales | West-251 |
| 1091 | n378o313p469 | rtran | Sales | Central-230 |
| 1092 | o391p779q935 | lpark | Sales | West-227 |
| 1098 | u671v146w618 | tarchamb | Sales | North-423 |
| 1099 | v283w690x104 | anaser | Finance | West-357 |
| 1105 | b551c837d758 | kmei | Finance | Central-232 |
| 1107 | d168e758f876 | akajwara | Sales | North-471 |
| 1109 | f229g533h679 | nlocklea | Sales | East-196 |
| 1110 | g567h376i314 | pchaudhu | Sales | Central-428 |
| 1111 | h835i179j862 | jlee | Sales | West-309 |
| 1116 | m272n572o874 | nzhao | Sales | South-100 |
| 1117 | n683o758p820 | dahmad | Sales | West-405 |
| 1118 | o305p208q337 | jpark3 | Sales | South-329 |
| 1119 | p164q780r999 | omubarak | Sales | West-409 |
| 1121 | r628s557t397 | mrojas | Sales | East-288 |
| 1122 | s103t952u851 | btorres | Finance | West-319 |
| 1130 | a317b635c465 | tsnow | Sales | Central-451 |
| 1136 | g299h520i457 | jhawes | Finance | West-416 |
| 1138 | i671j355k725 | sromero | Finance | South-329 |
| 1142 | m674n127o823 | lsilva | Finance | East-440 |
| 1144 | NULL | erobinso | Finance | Central-266 |
| 1147 | r454s225t299 | tvega | Finance | West-177 |
| 1148 | s328t505u907 | dharvey | Finance | South-181 |
| 1159 | d881e710f732 | jshen | Finance | East-193 |
| 1164 | i682j513k442 | fsmeltz | Finance | North-163 |
| 1169 | NULL | mmitchel | Sales | Central-250 |
| 1174 | s371t911u987 | eortiz | Finance | North-428 |
| 1175 | t959u687v394 | jclark2 | Finance | North-194 |
| 1176 | u849v569w521 | nliu | Sales | West-220 |
| 1181 | z803a233b718 | sessa | Finance | South-207 |
| 1185 | d790e839f461 | revens | Sales | North-330 |
| 1186 | e281f433g404 | sacosta | Sales | North-460 |
| 1187 | f963g637h851 | bbode | Finance | East-351 |
| 1188 | g164h566i795 | noshiro | Finance | West-252 |
| 1195 | n516o853p957 | orainier | Finance | East-346 |

71 rows in set (0.001 sec)

MariaDB [organization]>

**Explanation:**
The `OR` operator returns records where either condition is true:
- `department = 'Finance'` retrieves Finance department employees
- `department = 'Sales'` retrieves Sales department employees

Using `OR` ensures employees from both departments are captured in a single query without running two separate queries.

---

## Retrieve All Employees Not in IT

A final security update was needed for all employees outside of the Information Technology department. I used the following query to exclude IT employees.

```sql
SELECT *
FROM employees
WHERE NOT department = 'Information Technology';
```

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE NOT department = 'Information Technology';
+-------------+-------------+----------+-----------------+-------------+
| employee_id | device_id   | username | department      | office      |
+-------------+-------------+----------+-----------------+-------------+
|        1000 | a320b137c219 | elarson  | Marketing       | East-170    |
|        1001 | b239c825d303 | bmoreno  | Marketing       | Central-276 |
|        1002 | c116d593e558 | tshah    | Human Resources | North-434   |
|        1003 | d394e816f943 | sgilmore | Finance         | South-153   |
|        1004 | e218f877g788 | eraab    | Human Resources | South-127   |
|        1005 | f551g340h864 | gesparza | Human Resources | South-366   |
|        1007 | h174i497j413 | wjaffrey | Finance         | North-406   |
|        1008 | i858j583k571 | abernard | Finance         | South-170   |
|        1009 | NULL         | lrodriqu | Sales           | South-134   |
|        1010 | k242l212m542 | jlansky  | Finance         | South-109   |
|        1011 | l748m120n401 | drosas   | Sales           | South-292   |
|        1015 | p611q262r945 | jsoto    | Finance         | North-271   |
|        1016 | q793r736s288 | sbaelish | Human Resources | North-229   |
|        1017 | r550s824t230 | jclark   | Finance         | North-188   |
|        1018 | s310t540u653 | abellmas | Finance         | North-403   |
|        1020 | u899v381w363 | arutley  | Marketing       | South-351   |
|        1022 | w237x430y567 | arusso   | Finance         | West-465    |
|        1024 | y976z753a267 | iuduike  | Sales           | South-215   |
|        1025 | z381a365b233 | jhill    | Sales           | North-115   |
|        1026 | a998b568c863 | apatel   | Human Resources | West-320    |
|        1027 | b806c503d354 | mrah     | Marketing       | West-246    |
|        1028 | c603d749e374 | aestrada | Human Resources | West-121    |
|        1029 | d336e475f676 | ivelasco | Finance         | East-156    |
|        1030 | e391f189g913 | mabadi   | Marketing       | West-375    |
|        1031 | f419g188h578 | dkot     | Marketing       | West-408    |
|        1034 | i679j565k940 | bsand    | Human Resources | East-484    |
|        1035 | j236k3031245 | bisles   | Sales           | South-171   |
|        1036 | k5501533m205 | rjensen  | Marketing       | Central-239 |
|        1038 | m873n636o225 | btang    | Human Resources | Central-260 |
|        1039 | n253o917p623 | cjackson | Sales           | East-378    |
|        1040 | o783p832q294 | dtarly   | Human Resources | East-237    |
|        1041 | p929q222r778 | cgriffin | Sales           | North-208   |
|        1042 | q175r338s833 | acook    | Human Resources | West-381    |
|        1044 | s429t157u159 | tbarnes  | Finance         | West-415    |
|        1045 | t567u844v434 | pwashing | Finance         | East-115    |
|        1046 | u429v921w138 | daquino  | Finance         | West-280    |
|        1047 | v109w587x644 | cward    | Finance         | West-373    |
|        1048 | w167x592y375 | tmitchel | Finance         | South-288   |
|        1049 | NULL         | jreckley | Finance         | Central-295 |
|        1050 | y132z930a114 | csimmons | Finance         | North-468   |
```

**Explanation:**
- `NOT` negates the condition, excluding all records that match
- `department = 'Information Technology'` identifies IT employees
- The result returns every employee in all other departments

This ensures the security update is applied to all non-IT staff.

---

## Summary

In this activity, I used SQL filters to investigate security issues across two tables — `log_in_attempts` and `employees`. I applied `AND` to narrow results by combining multiple conditions, `OR` to retrieve records matching either of two conditions, and `NOT` to exclude specific records from results. I also used the `LIKE` operator with the `%` wildcard to match patterns in country and office fields. These queries enabled me to isolate suspicious login activity outside business hours, investigate logins on specific dates, identify login attempts from outside Mexico, and retrieve targeted employee groups for security updates across multiple departments.